

Xun (Vincent) Wang

(940) 312 - 9951 | XunVWang0123@gmail.com | Rapid City, South Dakota

Curriculum Vitae

Education

Ph.D. in Mechanical and Energy Engineering , GPA: 3.96/4	01/2018 - 12/2024
University of North Texas	Denton, Texas, U.S.A.
M.S. in Engineering Technology, Mechanical Systems , GPA 3.75/4	08/2015 - 12/2017
University of North Texas	Denton, Texas, U.S.A.
M.S. in Long-Term Care, Senior Housing and Aging Services , GPA 3.83/4	08/2010 - 05/2014
University of North Texas	Denton, Texas, U.S.A.
B.S. in Mechanical Engineering and Automation	08/2005 - 05/2009
Beijing Institute of Graphic Communication	Beijing, China

Experience

Assistant Professor	08/2026 - Present
Lecturer	08/2025 - 08/2026
Adjunct Lecturer	01/2025 - 08/2025

Leslie A. Rose Department of Mechanical Engineering, South Dakota School of Mines and Technology
Rapid City, South Dakota, U.S.A.

- Established the thermo-fluid science research laboratory.
- Mentored undergraduate and graduate students in research activities.
- Coordinated courses and developed curricula.
- Provided academic advising.
- Advised senior design projects.
- Prepared class materials and delivered lectures for mechanical engineering fundamental courses.
- Demonstrated the operations of machines in the manufacturing shop.
- Supervised students using machines and ensured their safety.
- Assessed student performance.
- Course list:
 - ME 126/L Design for Manufacturing/Lab (Course Coordinator)
 - ME 211 Introduction to Thermodynamics (Course Coordinator)
 - ME 312 Thermodynamics II (Course Coordinator)
 - ME 404 Heating, Ventilation, and Air Conditioning

Teaching Assistant

01/2018 - 12/2024

Department of Mechanical Engineering, University of North Texas

Denton, Texas, U.S.A.

- Evaluated and graded assignments, quizzes, and exams for the following courses: Statics, Dynamics, Mechanics of Materials, Data Analysis, Finite Element Analysis, Biocomposites, and Mechanical Vibrations.
- Assisted instructors in posting announcements/assignments and delivering students' grades in a timely manner through the CANVAS system.
- Led lab sessions and prepared tutorials in Finite Element Analysis, Mechanical and Energy Engineering Laboratory II, and Biocomposites.
- Gave the lecture on Metrology and instructions of Lathe, Mill, Drill Press, Saws, Sand Casting, Arc Welding, Plasma Cutting, Sheet Metal Bending/Cutting, Soldering, and Brazing for the course of Manufacturing Processes.
- Assisted students in their Senior Design I & II projects.

Assisted Living Universal Caregiver

06/2014 - 04/2015

Intern

01/2014 - 06/2014

Good Samaritan Society - Lake Forest Village

Denton, Texas, U.S.A.

- Provided medical and daily service to assisted living residents.

Research**Doctoral Dissertation**

08/2019 - 12/2024

Bio-Based Fatty Acid Synthesization from Recycling Waste Cooking Oil and Its Applications

- Created experimental methods of producing waste cooking oil based fatty acids
- Employed hydrolysis and hydrogenation processes and produced the fatty acids with different melting temperatures and latent heat values
- Developed logarithmic models that describe the relationship between hydrogenation time, melting point, and latent heat
- Provided a novel method to recycle waste cooking oil for reducing reliance on synthetic phase change materials and supporting the development of a circular economy
- Wrote one journal paper (in submission)

Investigator for CDC NIOSH Project

07/2021 - 06/2023

Optimization for Thermal Protection of Firefighter's Glove by Phase Change Material

- Created, conducted, and optimized numerical simulations independently.
- Validated simulation results by sample fabrications and test data collections.
- Analyzed collected data and visualized results.
- Wrote project reports monthly and published two journal papers.

Side Projects***Phase Change Material Integrated Testing Chamber***

07/2023 - 06/2024

- Built 3 ft. × 3 ft. × 3 ft. phase change material integrated chamber.
- Assembled a cooling system and attached phase change material insulation boards to the chamber.
- Assembled heating and cooling systems to build a larger temperature-controlled environment.
- Tested the energy consumption for the chamber in the larger environment.

- Analyzed the energy consumption data and wrote reports.

Phase Change Material Embedded Chest Freezer

01/2020 - 12/2020

- Numerically simulated heat transfer and energy conservation under static and dynamic conditions.
- Built phase change material embedded freezer and conducted energy conservation test.
- Reported test results to project funder.

Housing Construction Board Fabricated from Recycled Denim Fiber

01/2018 - 12/2018

- Collaborated with colleagues to fabricate 4 ft. × 8 ft. boards.
- Tested mechanical properties of the boards.

Service

- **Reviewer**, American Society of Thermal and Fluids Engineers (ASTFE)
10th Thermal and Fluids Engineering Conference (TFEC) 10/2024
- **Committee Member**, Undergraduate Curriculum Committee
Leslie A. Rose Department of Mechanical Engineering, South Dakota School of Mines and Technology 08/2025 - Present

Patent

- Zhao, X., Wei, Q., Zhang, X., Zhang, M., Wang, X., Guo, X., Zhao, X. (2009). Wire Deicer: CN201204425Y.

Honors and Awards

- CENG/DT Award (85%) (\$500 each semester) 10/2019 - 03/2022
- TGS R.B. Toulouse Scholarship (Doctoral & MFA) (\$1,000) 10/2022
- CENG/DT Award (85%) (\$500 each semester) 10/2023 - 03/2024

Publications

- Wang, X., Fu, Y., Washburn, S., Nguyen, T. B., Chauhan, A., Shaikh, S., Zhao, W., Wang, H., Shi, S. Q. (2026). Fatty Acid from Recycled Waste Cooking Oil as Phase Change Material for Thermal Storage System, *RSC Advances*, 16(7), 5852-5863. <https://doi.org/10.1039/D5RA07469J>.
- Wang, X., Zhao, W., Xu, S.S. Pollard, J. (2025). Effects of moisture contents on temperature control performance of phase change material integrated structural firefighting gloves, *Fire Safety Journal*, 56, 104444. <https://doi.org/10.1016/j.firesaf.2025.104444>.
- Smith, L. M., Fu, Y., Pittala, R. K., Wang, X., Jabel, C., Masignag, K., Arellanes, J., Ghosh, M., Shi, S. Q., Ecker, M., Wang, C. (2025). Bacteria Retted Hemp Fiber/PLA Composites. *Processes*, (13)4, 1000. <https://doi.org/10.3390/pr13041000>.
- Wang, X. (2024). Bio-Based Fatty Acid Synthesization from Recycling Waste Cooking Oil and Its Applications. University of North Texas. <https://oi.10.12794/metadc2415974>.
- Wang, X., Zhao, W., Pollard, J., Xu, S.S. (2024). Experimental study on the thermal protection enhancement of novel phase change material integrated structural firefighting gloves under high-heat exposures, *Case Studies in Thermal Engineering*, 56, 104286. <https://doi.org/10.1016/j.csite.2024.104286>.

- Huang, Z., Wang, X., Cai, L., Tao, Y., Tolone, W. J., El-Shambakey, M., Bhattacharjee, S. D., & Cho, I. (2020). Blast risk assessment of wood residential buildings: West fertilizer plant explosion case. *Journal of performance of constructed facilities*, (34)3, 04020022. [https://doi.org/10.1061/\(ASCE\)CF.1943-5509.0001414](https://doi.org/10.1061/(ASCE)CF.1943-5509.0001414).
- Nasrazadani, S., Sekar, S., Padamati, N., Wang, X., & Gamez-Rivas, F. (2019). Training Graduate Students in Utilization of Analytical Instruments in a Failure Analysis Course. *2018 Gulf Southwest Section Conference*. <http://dx.doi.org/10.26153/tsw/6943>.

Presentations

- Wang, X., Zhao, W., Xu, S. S. The 9th Thermal and Fluids Engineering Conference (TFEC), Experimental Study of Phase Change Material Integrated Firefighters' Gloves under Moisture Conditions, Corvallis, Oregon, April 21-24, 2024. (Oral presenter).
- Wang, X., Shi, S. Q., Zhao, W. The 63rd Society of Wood Science and Technology (SWST) International Convention, Thermal Performance Test of Cellulose Insulation Imbedded with Phase Change Material, Virtual Conference due to COVID, formerly Portorož, Slovenia, July 12-15, 2020. (Poster Presenter).

Professional Membership

- | | |
|--|-------------------|
| • American Society of Thermal and Fluids Engineers (ASTFE) | 01/2024 - Present |
| • Society of Wood Science and Technology (SWST) | 2020/01 - 2021/12 |